

Classification of Arc Fault in Sphere-Gap and Rod-Gap Using Stockwell Transform and Machine Learning Based Approach

Himadri Lala* and Subrata Karmakar
Department of Electrical Engineering
National Institute of Technology, Rourkela
Rourkela-769008, India
E-mail: lalacemk@gmail.com*

Abstract— There are different types of arc may occur depending on the arcing conditions and involved surfaces. The severity of the arc is determined by the involved arcing surface and the arcing current flowing path. In this study, arc in Sphere-Gap and Rod-Gap surfaces is considered for the time-frequency domain analysis. The voltage characteristics for both the arc events are recorded in different voltage levels and gap length. A Stockwell Transform (ST) based approach is applied on the arc signals for the harmonic decomposition. Further, K-Nearest Neighbor (KNN) machine learning algorithm is applied on the ST coefficients for the classification of real-time arc signals of different arcing conditions. The results obtained using ST and KNN algorithm successfully classifies different arc faults due to rod-gap and sphere-gap by their harmonic signature.

Keywords — arc discharges, Stockwell transform, fault diagnosis, high impedance arc fault, machine learning

I. INTRODUCTION

A high impedance arc fault (HIAF) leaves the energized conductor open on the surface which can pose a serious threat to the living beings. However, in practice, the conventional fault detection equipment fails to detect high impedance arc due to its minor impact on the current profile and almost all the equipment works on over-current principle [1], [2]. Moreover, the value of the fault current can be less than one-tenth of the load current. It appears to be harmless but, it can easily cause fire because of the arc flash heating which may rise to thousands of degree Celsius over a long period time. On the other hand, an electrical arc may also occur due to the fault in the equipment, insulation or between two adjacent conductors. Moreover, the arcing surface is the primary factor for analyzing different arc faults.

Over the years, many studies have been carried out on high impedance fault (HIF), and it has been observed that the voltage and current waveform in the presence of HIF are random in nature due to the non-linearity in fault impedance. However, many techniques such as power line carrier communication [3], mathematical morphology [4], [5], magnetic field signature based analysis [6] are used to detect the HIF. On the other hand, harmonic analysis techniques [7] are also used for the HIF detection. Scores of researchers have adopted discrete wavelet transform (DWT) along with machine learning techniques [8], [9], such as the probabilistic neural network (PNN), artificial neural network (ANN), support vector machine (SVM), etc. for the HIF detection.

Choi-Williams distribution (CWD) is also used along with time-frequency distribution (TFD) for HIF detection [10]. Apart from the simulations in [11] simulation as well as experimental verification of high impedance arc characteristics is described. Here, a DWT based detection technique is also proposed for arc due to the tree leaning on the power conductor. Nevertheless, the computational difficulty in frequency domain analyses motivated few researchers [12] to find the solution for arc detection even in time-domain. The literature related to arc fault has only discussed the features of high impedance arc faults with non-linear grounding paths i.e. tree-line or line-surface, but, sometimes grounding path for an arc fault may not be a high impedance or nonlinear one. The arc may occur in any places which can hold an equipment failure such as panel-boards, metal clad switchgear, switchboards, motor control centers, motor starters and drive cabinets and transformers. On the other hand, an electrical arc may also occur due to the fault in the equipment, insulation or between two adjacent conductors. The fundamental difference of arc in leaning-tree or other non-linear surfaces (non-metallic) and electrical equipment or conductor (metallic) is the grounding path for the fault current, which depends on grounding impedances. Moreover, the arcing surface is the primary factor for analyzing different arc faults.

Therefore, the classification of arc in metallic surfaces is necessary to identify the type of involved arcing surfaces and improve the protection scheme of a distribution system. The primary objective of the work is to detect and classify high impedance arc faults in rod-gap and sphere-gap.

II. ARC FAULT CLASSIFICATION ALGORITHM

In an arc flash, a large amount of concentrated radiant energy dissipates outward from electrical hardware in form of an arc. The arc makes high-pressure waves which can be dangerous to the environment as well as to the living beings around it by causing fire. A Stockwell Transform (ST) and K-Nearest Neighbor (KNN) based algorithm is developed for detection and classification of high impedance arc in rod-gap and sphere-gap.

A. Application of S-Transform and feature extraction

In a given time-series of a fault signal $f(t)$, the local spectrum at time $t = \tau$ can be obtained by multiplying $f(t)$ with a Gaussian function $g(t)$ located at $t = \tau$. The 'Stockwell Transform' used in this study is given in Eqn. (1).

$$S(f, \tau, \sigma) = \int_{-\infty}^{\infty} f(t)g(t - \tau)e^{-i2\pi ft} dt \quad (1)$$

where, $g(t) = \frac{|f|}{\sqrt{2\pi}} e^{-\frac{t^2 f^2}{2}}$

Here, σ represents the width of the Gaussian function $g(t)$. The ST of a fault signal $f(t)$ is defined as a CWT with a specific mother wavelet multiplied by the phase factor $p_1(t, f)$.

$$p_1(t, f) = f(t)e^{-i2\pi ft} \quad (2)$$

Substituting in the Eqn (8),

$$S(f, \tau, \sigma) = \int_{-\infty}^{\infty} p_1(t, f)g(t - \tau)dt \quad (3)$$

$$= p_1(t, f) * g(t, \sigma) \quad (4)$$

Here, '*' denotes the convolution operation. The ST provides a time-frequency representation instead of the time scale representation developed by the WT [13].

B. Arc fault classification using KNN

Different type KNN based machine learning algorithms are used for classification of arc faults in different arcing surfaces. Among different machine learning algorithms for classification, one of the promising machine learning technique that is widely used is KNN. In k-NN classification, the output is a class membership. An object is classified by a majority vote of its neighbors, with the object being assigned to the class most common among its k nearest neighbors (k is a positive integer, typically small) [14]. If $k = 1$, then the object is simply assigned to the class of that single nearest neighbor. It is one of the simplest forms of machine learning. Therefore, it requires less computational requirement. The complete methodological representation of the proposed classification algorithm using KNN and cross-validation is shown in Fig. 1.

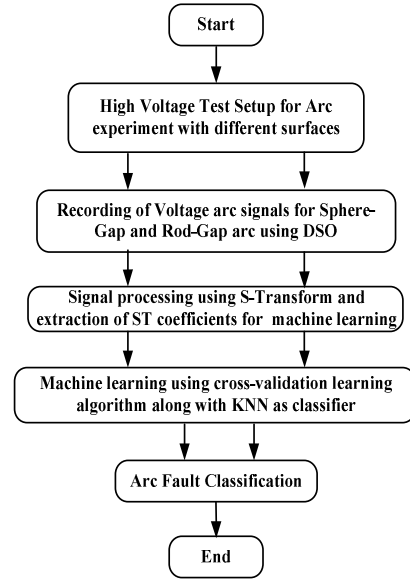


Fig. 1. Flowchart for HIAF detection

III. ARRANGEMENTS FOR SPHERE-GAP AND ROD-GAP ARC EXPERIMENT

It is very difficult to simulate arcing phenomena in a real-time environment due to the equipment cost and the risk associated with it. In the high voltage engineering laboratory at National Institute of Technology, Rourkela, India, a 500 kV, 500 kVA power frequency testing transformer (Make-W. S. Test Systems; Type-AC 500-500) is used for the experiment of different arc fault involving conductor and concrete, grass, tree and wet-sand as arcing surface/medium. It is a remotely operated HV laboratory with a capacity for experiments up to 500 kV level which includes extensive testing and detection of partial discharge (PD) with an advanced virtual instrumentation based research facility [15], impulse, arcing, lightning phenomena and insulation failure etc. The rated input/output voltage for the testing transformer is 610 V/500 kV, 50 Hz and the rated test voltage is 450 kV.

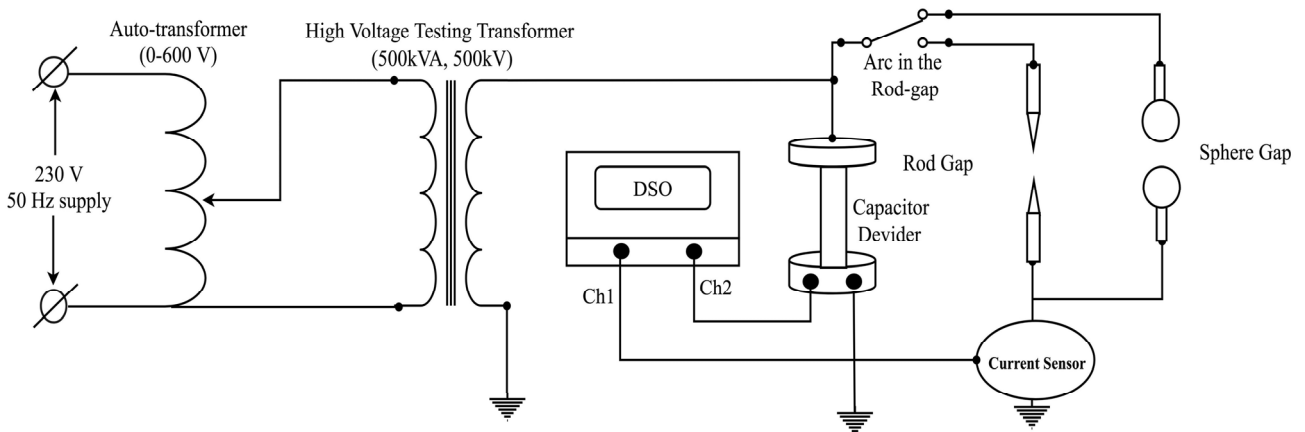


Fig. 2. Schematic diagram of the overall experimental setup for different arcing conditions

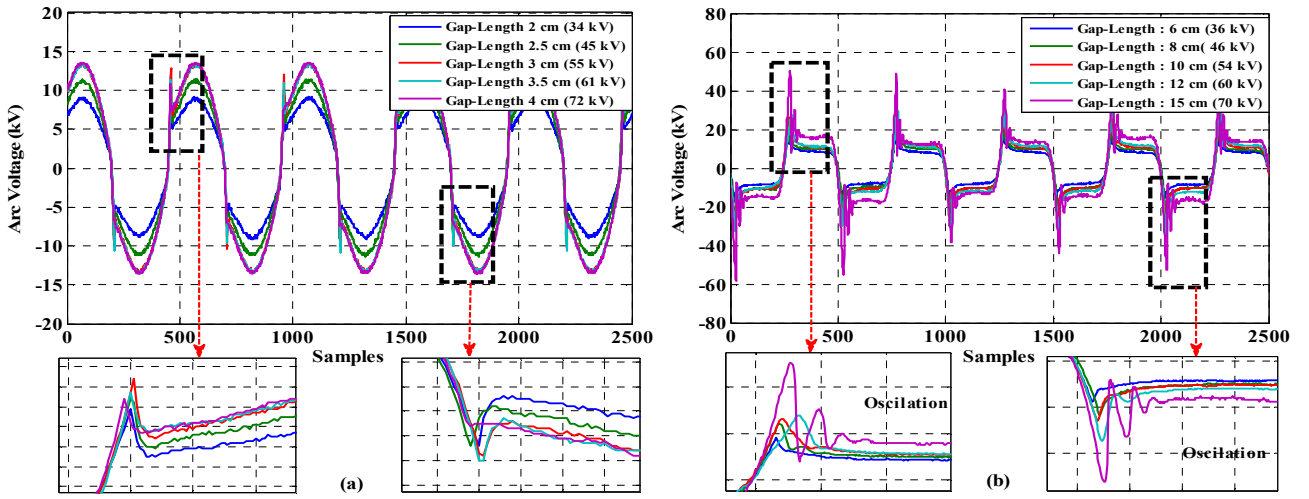


Fig. 3. Voltage signals captured during the arcing experiment between metallic surfaces for different gap length and arc voltage; (a) Sphere-gap arc, (b) Rod-gap arc

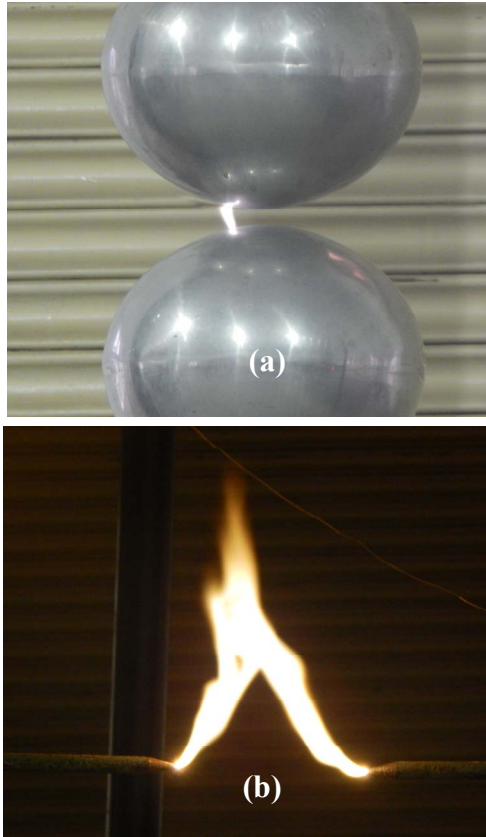


Fig. 4. Arcing phenomena in different arcing surface; (a) Sphere-gap, (b) Rod-gap

TABLE I. SPECIFICATION OF THE ARCING EXPERIMENT EQUIPMENT

Sl.	Description	Ratings
1	HV Test Transformer	Rated Test Voltage:0-500 kV Rated output power:500 kVA, 250 mA, (125 kVA) cont. duty 500 kVA,1 A (500 kVA) 5 min ON; Rated PD level: less than 10 pC at rated voltage
2	Control Desk with oil cooled auto-transformer	320 kVA; Input: 2 Ph 415 V AC 50 Hz; Output: 0-610 V/ 420 A
3	Capacitor Measuring	500 kV, 1000 pF PD < 5 pC
4	Voltmeter	(AC peak--- 4.5 Digit, Digital Display)
5	Compensating Reactor	0-600 V/50 Hz AC; 50/100/150/200 and 250 kVAR with common tapping C

The Fig. 2 explains the schematic diagram for different arcing experimental arrangement. Here, an auto-transformer is used as an input to the testing transformer. The auto-transformer gives a voltage of 0-610 V to the low voltage side of the high-voltage testing transformer. The voltage from the testing transformer is applied to the conductor through capacitor divider and PT. The voltage signals captured during the arcing experiment in metallic surfaces i.e. Sphere-gap and Rod-Gap for different gap length and arc voltage is shown in Fig. 3a and 3b respectively. In Fig. 4a and 4b different type of arc during different experiments are shown. The applied voltage to the conductor and different arcing surface (i.e. sphere-gap and rod-gap) is measured through a PT of 1000:1 ratio. The rated output power of the HV test setup is 500 kVA, 250 mA, (125 kVA). The detailed specification and description is depicted in Table I. In this whole experiment, arc in sphere-gap and arc in rod-gap is considered.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The V-I characteristics of arc fault can vary depending on the nature of arc. The time constant of an arc determines the V-I characteristics. Nevertheless, several arc characteristics may also change depending on other parameters. In this study, two types of arc are seen after two different sets of experiments for the analysis. The sphere-gap experiment is performed with different gap lengths (2, 2.5, 3, 3.5, 4 cm etc.). The sphere assembly which is used in this experiment is of 25 cm diameter. The arcing voltage measured for 2, 2.5, 3, 3.5, 4 cm sphere-gap length is 34, 45, 55, 61, 72 kV respectively. The sphere assembly used here can be found in Fig. 4a. The arc produced between sphere-gap always starts from the minimum distance point and moves along the curved surface by forming a conical pattern. The experiments are repeated, and a similar outcome has been observed. On the other hand, the rod gap experiment assembly used in this work is described in Fig. 4b. The rod-gap length is varied from 6 cm, 8 cm, 10 cm, 12 cm, 15 cm and beyond. The arc breakdown voltage found for different gap lengths are 36, 46, 54, 60, 70 kV respectively for different rod gaps mentioned above. The rods used to make the experimental setup are of 1.5 cm diameter. A needle-like shape has been given to the rods.

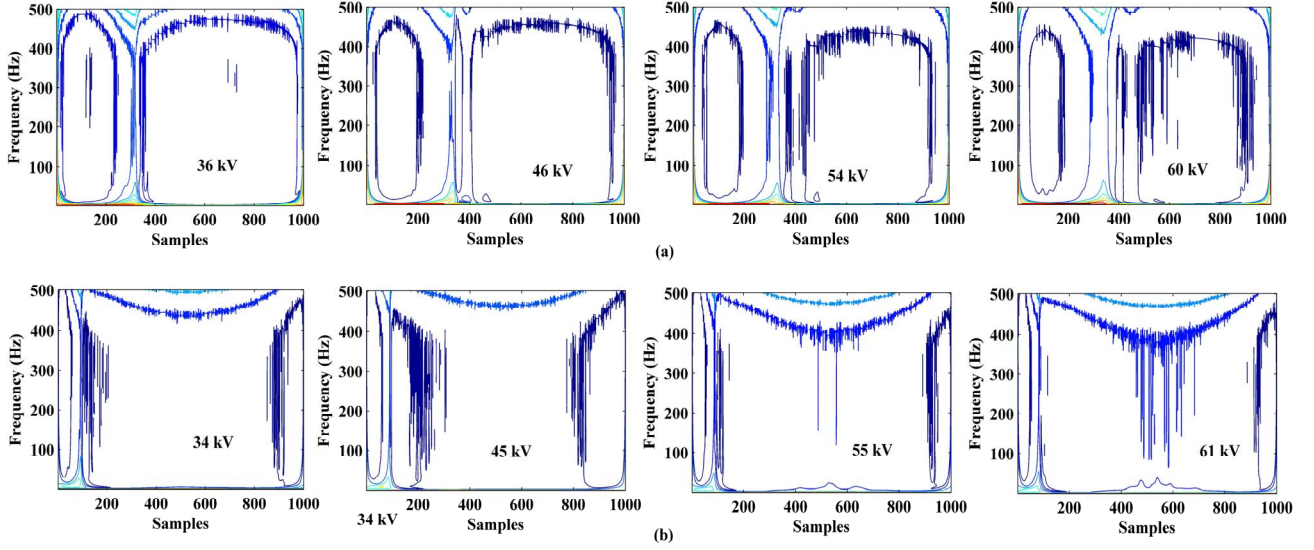


Fig. 5. Contour plot of ST coefficients for different arcing conditions (a) Rod-gap, (b) Sphere-gap

TABLE II. CLASSIFICATION RESULT FOR THE DIFFERENT TYPE OF ARC USING DIFFERENT KNN MACHINE LEARNING ALGORITHMS AND ST

Different Test Sets	Number of Input Feature: 501(ST coefficients) and Number of Class : 2 (Sphere-Gap and Rod-Gap)							
	Arcing voltage in the different surfaces (kV)		Classification accuracy using different classifiers (%)					
	Sphere Gap	Rod Gap	Fine KNN	Medium KNN	Coarse KNN	Cosine KNN	Cubic KNN	Weighted KNN
Test 1	34	36	100	100	98.1	99.3	100	100
Test 2	45	46	100	99.3	98.3	99.4	100	100
Test 3	55	54	100	100	99.3	100	100	100
Test 4	61	60	100	100	99.2	100	100	100
Test 5	72	70	100	100	97.4	100	100	100

During rod-gap experiment, it has been found that the arc between the rods starts from the shortest path between them and gradually takes a triangular shape and moves upward till its withstand limit. On the other hand, as the arcing voltage keeps on increasing with gap length in case of rod gap a damping oscillation is found. This kind of oscillation in transient is not visible in case of Sphere-Gap arc. This phenomenon can be seen in Fig. 3a and 3b. The experiments were repeated, and the similar outcome has been found. Therefore, the arcs in two different type of metallic surface exhibit different phenomena depending on their arcing surface. In this study, the Stockwell transform (ST) is used as a signal processing tool to analyze the arc signals. Further, the ST coefficients are used to extract the different features from the transient. Here, the ST coefficients are extracted and used for detection of the high impedance arc. ST gives detailed time-frequency coefficients when applied on a signal. In this study, the ST is applied on the arc signals to get a comparative view of the machine learning results. Firstly, ST is applied and the coefficients are calculated. As the ST gives much more detailed coefficients, a window of 1000 samples of different arc signals are used for the analysis. Further, the calculated ST coefficients are utilised to build the training sets for the machine learning algorithms. Here, the ST coefficients are used as the input to the machine learning algorithms (i.e. Fine KNN, Medium KNN, Cubic KNN, Cosine KNN and Weighted KNN). In case of ST, the number of feature calculated from arc signals is 501. The contour plot of ST

coefficients for both the sphere-gap and rod-gap arc at different arcing voltage is shown in Fig. 5. It is clearly evident from Fig. 5a and 5b that both the arcing surfaces gives different type of harmonic signature. The results for both the different learning algorithm are recorded using cross verification. In this work, the classification between sphere-gap and rod-gap arc is performed. Here, two classes correspond to sphere-gap and rod-gap. Table 2 illustrates the results recorded after both the learning algorithms applied. The percentage of accuracy in classification shows the efficiency of different KNN algorithm. In this context, Fine KNN, Medium KNN, Cubic KNN and Weighted KNN are performing reasonably good to classify sphere-gap and rod-gap arcs. The results obtained using ST-KNN based classification algorithm clearly shows the difference in harmonic profile of both the arcing phenomena. However, the results calculated using ST coefficients are found to be reasonable for most of the KNN based techniques.

V. CONCLUSION

The results obtained from both the experiments of sphere-gap and rod-gap clearly show the differences in both time and time-frequency domain. In time domain, the voltage characteristics for sphere-gap and rod-gap is different from each other. On the other hand, contour plots found after applying ST also supports the claims of time-domain. The time-frequency domain analysis using ST successfully classifies the two events on the basis of harmonic signature.

The main conclusions of the work are as follows:

- Depending on the grounding path or the arcing surface, arc fault can be of different types.
- ST along with different KNN successfully classifies the HIAF for sphere-gap and rod-gap with a reasonable level of classification accuracy.

The future research is planned with an emphasis on the variety of other arcing surfaces i.e. different kind of soil, etc. and the change in arcing response. However, the different type of non-linearity offered by other surfaces is also the future prospect of the work.

REFERENCES

- [1] J. C. Chen, B. T. Phung, D. M. Zhang, T. Blackburn, and E. Ambikairajah, "Study on high impedance fault arcing current characteristics," in *2013 Australasian Universities Power Engineering Conference (AUPEC)*, 2013, pp. 1–6.
- [2] J. Chen, T. Phung, T. Blackburn, E. Ambikairajah, and D. Zhang, "Detection of high impedance faults using current transformers for sensing and identification based on features extracted using wavelet transform," *Transm. Distrib. IET Gener.*, vol. 10, no. 12, pp. 2990–2998, 2016.
- [3] A. Ghaderi, H. L. Ginn, and H. A. Mohammadpour, "High impedance fault detection: A review," *Electric Power Systems Research*, vol. 143, pp. 376–388, 2017.
- [4] P. K. Routray, P. Mishra, M. Rout, "High Impedance Fault Detection in Radial Distribution System Using S-Transform And Neural Network," *2015 IEEE Power, Commun. Inf. Technol. Conf.*, pp. 545–551, 2015.
- [5] A. N. Milioudis, G. T. Andreou, and D. P. Labridis, "Enhanced protection scheme for smart grids using power line communications techniques-Part II: Location of high impedance fault position," *IEEE Trans. Smart Grid*, vol. 3, no. 4, pp. 1631–1640, 2012.
- [6] S. Gautam and S. M. Brahma, "Detection of High Impedance Fault in Power Distribution Systems Using Mathematical Morphology," *IEEE Trans. Power Syst.*, vol. 28, no. 2, pp. 1–1, 2012.
- [7] S. Gautam and S. M. Brahma, "Application of mathematical morphology based filters to detect a power swing," in *IEEE PES General Meeting, PES 2010*, 2010.
- [8] M. Sarlak and S. M. Shahrtash, "High-impedance faulted branch identification using magnetic-field signature analysis," *IEEE Trans. Power Deliv.*, vol. 28, no. 1, pp. 67–74, 2013.
- [9] D. C. Yu and S. H. Khan, "Adaptive high and low impedance fault detection method," *IEEE Trans. Power Deliv.*, vol. 9, no. 4, pp. 1812–1818, 1994.
- [10] T. M. Lai, L. A. Snider, E. Lo, and D. Sutanto, "High-impedance fault detection using discrete wavelet transform and frequency range and RMS conversion," *IEEE Trans. Power Deliv.*, vol. 20, no. 1, pp. 397–407, 2005.
- [11] H. Livani and C. Y. Evrenosoglu, "A Fault Classification and Localization Method for Three-Terminal Circuits Using Machine Learning," *Power Delivery, IEEE Transactions on*, vol. 28, no. 4, pp. 2282–2290, 2013.
- [12] A. Ghaderi, H. Mohammadpour, H. Ginn III, and Y.-J. Shin, "High Impedance Fault Detection in Distribution Network using Time-Frequency Based Algorithm," *IEEE Trans. Power Deliv.*, vol. 8977, no. c, pp. 1–1, 2014.
- [13] R. G. Stockwell, L. Mansinha, and R. P. Lowe, "Localization of the complex spectrum: The S transform," *IEEE Trans. Signal Process.*, vol. 44, no. 4, pp. 998–1001, 1996.
- [14] S. Tan, "Neighbor-weighted K-nearest neighbor for unbalanced text corpus," *Expert Syst. Appl.*, vol. 28, no. 4, pp. 667–671, 2005.
- [15] S. Karmakar, "Virtual-Instrument-Based Online Monitoring System for Hands-On Laboratory Experiment of Partial Discharges," *IEEE Trans. Educ.*, vol. 60, no. 1, pp. 29–37, 2017.